

:::: Scope of Work Document

Scope of Work (SOW) - Is an agreement between an employee/contractor and employer outlining services work to be completed by the employee. SOW's are used commonly in technology jobs and contract work.

Please submit your Scope of Work Document detailing the project you plan to complete. You will also include your SOW with your Final Project PowerPoint/Slideshow)

Student Project – Scope of Work (SOW)

Project Title: Cohort Manager Pro (Roster · Attendance · Grades)

Student Name: Gari'un Godfrey

Date: 10/13/2025

1. Project Overview

(Write 2–3 sentences about what your project will do.)

A command-line C++ application to manage a small class/cohort. The program loads a roster from a file, lets the user record attendance (check-in/check-out) and assignment grades, and generates daily/overall reports to disk. It supports search (binary search by ID), sorting (by name/ID/average), and basic statistics (mean/median/high/low). Data persists in simple CSV text files.

2. Goals

- Implement a `Student` class with: `id`, `name`, `presentToday`, `checkInTime`, `checkOutTime`, `vector<int> grades`.

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File I/O:

- Load roster from `roster.csv`.
- Append attendance to `attendance_log.csv`.
- Save gradebook to `grades.csv`.

Menu actions (functions only—no external libraries):

1. List students (paged).
 2. Search by ID using binary search (requires ID-sorted view).
 3. Sort by name, ID, or average grade (implement your own simple sort—selection/insert).
 4. Mark check-in/check-out (timestamps).
 5. Add grades; compute average/median per student.
 6. Generate reports: `report_daily.txt` and `report_overall.txt`.
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3. Deliverables

- Program files (main.cpp + others)
 - Output.pdf with screenshots and explanation
 - GitHub submission with correct file structure
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4. Timeline

- Week 1–2: Define idea & set up GitHub
 - Week 3–4: Build program features
 - Week 5: Test & debug
 - Week 6: Write Output.pdf
 - Week 7: Submit final project
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5. Success Criteria

- The program compiles and runs with no crashes.
- Roster loads, attendance, and grades can be entered and persisted.
- Binary search by ID works on an ID-sorted view; sorts produce the correct order.
- Reports show accurate stats and are written to files.
- All required files are submitted on time with a clear structure.